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UNIVERSAL INTERLOCKING CLOSURE DEVICE

Abstract

An interlocking device that includes a track, the track including a sleeve, a wall gap extending longitudinally along a first side of the sleeve, and an interlocking component extending longitudinally along a second side of the sleeve with the second side being opposite the first side.

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Background/Summary

BACKGROUND

[0001] Despite the numerous existing types of interlocking closure systems, such as zippers, available or known to be used on various materials and for various applications, each interlocking closure system requires complicated installation. The complicated installation process essentially prevents replacing a portion of an interlocking closure system, which in turn can render the entire

interlocking closure system inoperable should a portion of the system become damaged. Because the interlocking closure systems are typically integral components of the device in which they are installed, often times the entire device must be replaced when a portion of the interlocking closure system is damaged. Additionally, the complicated installation of known interlocking closure systems prevents the ability to quickly add an interlocking closure system to a device that is not originally designed with an interlocking enclosure device. Therefore, additional improvements to existing interlocking closure systems are desired.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] The Detailed Description is set forth with reference to the accompanying figures. In the figures, the left-most digit(s) of a reference number identifies the figure in which the reference number first appears. The use of the same reference numbers in different figures indicates similar or identical items. Furthermore, the drawings may be considered as providing an approximate depiction of the relative sizes of the individual components within individual figures. However, the drawings are not to scale, and the relative sizes of the individual components, both within individual figures and between the different figures, may vary from what is depicted. In particular, some of the figures may depict components as a certain size or shape, while other figures may depict the same components on a larger scale or differently shaped for the sake of clarity.

[0003] FIG. 1A illustrates a top-front perspective view and an end view of an embodiment of a universal interlocking closure device/system.

[0004] FIG. 1B illustrates a top-front perspective view and an end view of an embodiment of a universal interlocking closure device/system.

[0005] FIG. 1C illustrates a top-front perspective view of an embodiment of a universal interlocking closure device/system, and/or an embodiment of a portion of an interlocking closure device/system.

[0006] FIG. 2 illustrates a front perspective view and an end view of an embodiment of a universal interlocking closure device/system with electrical circuit components incorporated therein, note the device/system is merely displayed in a curved position to illustrate the flexibility, as discussed hereinafter.

[0007] FIG. 3 illustrates a bottom-front view of an embodiment of a universal interlocking closure device/system and an end view of a sleeve engagement piece.

[0008] FIG. 4 illustrates a bottom-front perspective view of an example installation of an embodiment of an interlocking closure device/system component being installed on an existing interlocking closure system that sustained damage.

[0009] FIG. 5 illustrates a bottom-front perspective view of an embodiment of a single interlocking closure device configured to be coupled to a plurality of surfaces.

[0010] FIG. 6 illustrates a top-front perspective view and an end view of an embodiment of a universal interlocking closure device/system.

[0011] FIG. 7 illustrates a flow diagram of an example method associated with installation of a universal interlocking closure device/system.

DETAILED DESCRIPTION

Overview

[0012] This disclosure is directed to a universal interlocking closure system and/or device. More specifically, embodiments of an interlocking closure track that may be claimed as: 1) variations of a universal interlocking closure device that may replace a malfunctioning individual component of an existing interlocking closure device system; 2) variations of a universal interlocking closure device that may be combined with another universal interlocking closure device to replace a

malfunctioning interlocking closure system; 3) variations of a universal interlocking closure device that may allow for easily installing an interlocking closure device where an interlocking closure device was not previously installed; or 4) variations of a universal closure device that may be combined with another universal interlocking closure device to install an interlocking closure system where there was no previously installed interlocking closure system. Features are further described as shown in the figures and expressed in the claims.

[0013] Notably, the interlocking portions of the disclosed system may have some components similar to traditional interlocking zipper teeth, hook and loop structures, Ziploc®, or other interleaved binding elements/configurations.

[0014] In an embodiment, an interlocking closure track may include a sleeve. The sleeve may have a gap extending longitudinally along a first side of the sleeve. Additionally, an interlocking component extends longitudinally along a second side of the sleeve, where the second side is opposite the first side.

[0015] As used herein, the term “sleeve” may include hollow, continuously or non-continuously walled, tubular shapes, having cross-sections that are comparable to geometrical or what may be called quasi-geometrical shapes. For the sake of this disclosure, a quasi-geometrical shape may be defined as a shape which would be determined to be a nearly complete common shape, but for the fact that a gap exists in a side wall or in a portion of a side wall thereof. Thus, the side wall of the shape does not fully enclose the area or volume that would exist within the otherwise solid-walled shape. To be clear, where, in the cross-section the gap appears as merely a discontinuous line on a two-dimensional shape, in the three-dimensional sleeve, the gap may occur as a discontinuity that extends through the side wall along at least a portion of the length of the tubular shape. For example, a square is generally understood to have four equal sides, while a quasi-square may appear with a gap in a side wall thereof. Nevertheless, for the sake of convenience, unless stated otherwise, a geometric shape that has a gap in a side wall, will still be referred to as the name of the shape as is commonly used, rather than having to add the term “quasi” to each instance herein. That is, a “quasi-square” cross-sectional shape may be referred to and claimed as a “square” cross-section. Thus, hollow cylindrical shapes as well as other hollow, tubular shapes, i.e., tubular-walled structures having non-circular cross-sections, including rectangular, square, triangular, pentagonal, etc. cross-sections are contemplated as covered embodiments by the inventors of this disclosure.

[0016] Additionally, as used herein, the term “fabric” includes, but is not limited to, woven and non-woven textiles, fabrics, and other materials that may be produced using natural, synthetic, or semi-synthetic materials, metallic materials, rubber, plastic, or other suitable material.

Illustrative Embodiments

[0017] FIG. 1A illustrates a top-front perspective view of an embodiment of a universal interlocking closure system **100A** (hereinafter “interlocking system **100A**”) and an end view of the interlocking system **100A**. In an embodiment, the interlocking system **100A** may include a track **102(a)** (i.e., a first track) and a track **102(b)** (i.e., a second track), each of which may be used as a replacement for a component in a previously installed interlocking system and/or may be used for initial installation of an interlocking closure system device.

[0018] In an embodiment, the track **102(a)** may include a sleeve **104(a)** that has a first side and a second side opposite the first side. The sleeve **104(a)** may have a wall gap **106(a)** extending longitudinally along the first side of the sleeve **104(a)**. The sleeve **104(a)** may be fixedly secured to a strip of fabric **108(a)** having a first edge and a second edge opposite the first edge. The second side of the sleeve **104(a)** may be coupled to the first edge of the strip of fabric **108(a)**, and the strip of fabric **108(a)** may extend along a portion, or the entire length, of the second side of the sleeve **104(a)**. The second edge of the strip of fabric **108(a)** may be coupled to an interlocking component **110(a)**. The interlocking component **110(a)** may extend along a portion, or the entire length, of the second edge of the strip of fabric **108(a)**. The distance between the first edge of the strip of fabric **108(a)** may be any distance desired when the track **102(a)** is constructed. Although the sleeve

104(a) is shown as having a cylindrical shape in FIG. 1A, it is contemplated (and shown in other embodiments in FIG. 1B and FIG. 3) that sleeve **104(a)** may have any hollow tubular shape. [0019] Likewise, in an embodiment, the track **102(b)** may include a sleeve **104(b)** having a wall gap **106(b)**, a strip of fabric **108(b)**, and a corresponding interlocking component **110(b)**, which is oppositely configured to interlock with interlocking component **110(a)**, as shown for example, or with alternative interlocking elements.

[0020] In an embodiment, installation of the track **102(a)** and/or track **102(b)** is described in FIG. 7. In addition to the installation methods described in FIG. 7, installation and removal of the tracks **102(a)**, **102(b)** may be performed using a tool specifically designed for manipulating the tracks **102(a)**, **102(b)**, and/or other embodiments.

[0021] While it is contemplated that the sleeves **104(a)**, **104(b)** may be rigid or pliable, it is commonly known that most enclosure systems are relatively pliable for the purpose of conforming to the widely varying shapes and contexts in which they are used. Regardless, the sleeves **104(a)**, **104(b)** may be made of metal, plastic, rubber, fabric, or other suitable material, such as, but not limited to: a heat-shrink tubing; a shape-memory material; a material or composition of materials that permits longitudinal elastic flexure, while maintaining more rigidity through the width/thickness; or a metal with high ductility; etc. It is further contemplated there may be circumstances in which each of the sleeves **104(a)**, **104(b)** may be formed of different materials, respectively.

[0022] In an embodiment, the interlocking components **110(a)**, **110(b)** may include a row of teeth, respectively, for a zipper-style enclosure system, and may thus be used to replace an existing zipper system (e.g., a coat, a tent, an enclosure, etc.). For example, as is explained in more detail with respect to FIG. 7 below, a user may cover the existing respective rows of teeth in the zipper system being replaced by placing the original opposing zipper tracks within the sleeves **104(a)**, **104(b)**. Meaning specifically that the original opposing teeth are bound within respective tubular sleeves **104(a)**, **104(b)**, the existing material that secures the existing teeth is pinched in the gaps **106(a)**, **106(b)**, and the interlocking components **110(a)**, **110(b)** may function as respective replacement rows of teeth similar to, or distinct from, the teeth enveloped by the sleeves **104(a)**, **104(b)**.

[0023] In an embodiment, the interlocking system **100A** may also include a sliding joiner piece **112(a)**. The sliding joiner piece **112(a)** may be used to couple or de-couple the interlocking component **110(a)** and a corresponding interlocking component **110(b)**.

[0024] Additionally, and/or alternatively, as indicated above, different structural types of interlocking components (beyond the zipper teeth type) are contemplated for the interlocking components **110(a)** and **110(b)**, including, but not limited to: continuous interlocking systems (i.e., Ziploc® systems), hook and loop closure systems (i.e., Velcro systems), magnetic closure systems (e.g., opposing magnetic features), fastener systems that utilize buttons, clasps, toggles, or other suitable fasteners, etc. Moreover, the interlocking components **110(a)** and **110(b)** may include more than one type of interlocking component within a same track. For example, a first portion of the entire length of an interlocking component **110(a)** (or likewise interlocking component **110(b)**) may include a row of teeth, and a second portion may change to a part of a hook and loop system.

[0025] In light of the above, in some embodiments, a sliding joiner piece may be unnecessary to decouple the interlocking component **110** from another interlocking component. That is, though a zipper type of interlocking teeth graphic is depicted, as shown in FIG. 1C (description below), the box (pointed at by leader line **110(c)**) is intended to represent other types of closure systems, as one skilled in the art would understand. For example, if a track were configured to utilize a hook and loop closure system, an interlocking component would couple to a corresponding interlocking component that utilized a hook and loop system and decouple from the corresponding interlocking component that utilized a hook and loop system component without the use of the sliding joiner piece.

[0026] FIG. 1B illustrates a top-front perspective view of an embodiment of a universal

interlocking closure system **100B** (hereinafter “interlocking system **100B**”) and an end view thereof. While the structural elements appear similar to those of interlocking system **100A**, a noted distinction is the difference in the shape of the sleeves. That is, the sleeves **104(a)**, **104(b)** have circular cross-sectional shapes, while the sleeves of interlocking system **100B** are shown as having square cross-sectional shapes in FIG. **1B**. Thus, it is contemplated (as shown in other embodiments in FIG. **1A**, FIG. **1C**, and FIG. **3**) that the sleeves of an interlocking system may have any hollow tubular shape that is suitable to achieve the function intended. Further, although the embodiment shown in FIG. **1A** depicts both sleeves **104(a)** and **104(b)** having cylindrical shapes, respectively, it is also contemplated that a universal interlocking closure system may include tracks with sleeves of different respective shapes. For example, a first track with a circular-shaped sleeve and a second track with a square-shaped sleeve.

[0027] FIG. **1C** illustrates a top-front perspective view of an embodiment of a universal interlocking closure system **100C** (hereinafter “interlocking system **100C**”). The interlocking system **100C** may include a track **102(c)** that may be used as a replacement for a component in a previously installed interlocking system and/or may be used for initial installation of an interlocking closure system device.

[0028] In an embodiment, the track **102(c)** may include: a sleeve **104(c)** with a wall gap **106(c)**, a strip of fabric **108(c)**, and an interlocking component **110(c)**. As demonstrated in FIG. **1C** compared to FIG. **1B**, sleeve **104(c)** may have a cylindrical shape, a hollow tubular shape, or any shape suitable for effectively coupling to track to a desired location.

[0029] It is noted that FIG. **1C** intentionally depicts the interlocking component **110(c)** as a row of boxes. However, as indicated above, the boxes effectively may represent any interlocking zipper teeth consistent with a zipper-type interlocking system, as well as embodiments of an interlocking component utilizing hook and loop structures, Ziploc®, or other interleaved elements/configurations that create a desired binding feature.

[0030] FIG. **2** illustrates a front perspective view of an embodiment of a universal interlocking closure system **200** (hereinafter “interlocking system **200**”) of the instant disclosure. The interlocking system **200** may include a first track **202(a)** that may be used as a replacement for a component in a previously installed interlocking system and/or may be used for initial installation of an interlocking closure system device.

[0031] The tracks **202(a)**, **202(b)** may, respectively include: a sleeve **204(a)**, **204(b)**; a strip of fabric **206(a)**, **206(b)**; a circuit element **208(a)**, **208(b)**; a sheathing **207(a)** (a corresponding sheathing on track **202(b)** is not shown explicitly, but is understood to be similar to sheathing **207(a)**); and an interlocking component **210(a)**, **210(b)**. Either of tracks **202(a)**, **202(b)** may bear a sliding joiner piece **212** to join the interlocking components **210(a)**, **210(b)**.

[0032] It is contemplated that the circuit elements **208(a)**, **208(b)** may, respectively, include string LED circuits, exposed optical fibers, coated optical fibers (e.g., fiber optic cable, etc.), and/or power or other data carrying conductors embedded into or onto the strips of fabric **206(a)**, **206(b)**. Additionally, as demonstrated by the arrows in the end view of FIG. **2**, the circuit parts **208(a)**, **208(b)** may be located at any point along the strips of fabric **206(a)**, **206(b)** including at the sleeves **204(a)**, **204(b)**. Although FIG. **2** depicts the circuit elements **208(a)**, **208(b)** as identical, as indicated above, different elements may be used in each. Moreover, multiple circuit elements may be included in a single track. For example, one track may combine two or more of a light string, an audio cable, a power cable, or a data cable.

[0033] As seen in FIG. **2**, in an embodiment, the circuit elements **208(a)**, **208(b)** may include circuit connectors **214(a)**, **214(b)**, respectively. It is contemplated that the circuit connector **214(a)** and the circuit connector **214(b)** need not be identical. For example, the circuit element **208(a)** may include a fiber optic cable used for transmitting data and the circuit connector **214(a)** may include a connector common to connect fiber optic cables. In contrast, the circuit element **208(b)** may be an audio cable and the circuit connector **214(b)** may be an input jack/receptacle/receiver configured to

connect to an audio output device. Although FIG. 2 depicts circuit connectors **214(a)** and **214(b)** at one end of the circuit elements **208(a)**, **208(b)**, one or more circuit connectors may be installed at the opposite ends and/or other positions along each of the circuit elements (though not shown). [0034] Moreover, in an embodiment with sheathing, the sheathing **207(a)**, **207(b)** may surround the circuit elements **208(a)**, **208(b)**, and may be translucent or transparent such that light-emitting elements (e.g., a string light circuit of micro or mini LEDs) may emit light along a length or a portion thereof, as incorporated into the device/system/components. It is also contemplated that the sheathing **207(a)**, **207(b)** may be rigid or pliable and may be made of any material suitable to accomplish the functions desired. Additionally, and/or alternatively, the circuit elements **208(a)**, **208(b)** may include bare wire, coated wire, braided wire, fiber optic fibers, fiber optic cables, or other suitable power, current and/or data carrying conductors.

[0035] Additionally, as shown in FIG. 2, a user of the system may connect one or more of the circuit elements **208(a)**, **208(b)** to one or more controllers/subsystems **216**. For example, such a controller/subsystem **216** may include one or more of a heating controller, a lighting controller, a battery, an audio output/input device (e.g., headphones, phone, speakers, digital audio player, etc.); etc. The connection to the controller/subsystem **216** may allow the user to control a lighting scheme of the enclosure system, control a warming/heating element, control an audio element, etc. (if so equipped at the item/location for which the system has been installed).

[0036] FIG. 3 illustrates a bottom-front perspective view of an embodiment of a universal interlocking closure system **300** (hereinafter “interlocking system **300**”) of the instant disclosure and bottom-front perspective view of a sleeve engagement piece **306(c)**. The interlocking system **300** may include a first track **302(a)** that may be used as a replacement for a component in a previously installed interlocking system and/or may be used for initial installation of an interlocking closure system device. The interlocking system **300** may also include a material M. It is contemplated that material M may include woven and non-woven textiles, fabrics, other materials that may be produced using natural, synthetic, or semi-synthetic materials, metallic materials, rubber, plastic, or other suitable material.

[0037] The tracks **302(a)**, **302(b)** may respectively include: a sleeve **304(a)**, **304(b)**; a sleeve engagement piece **306(a)**, **306(b)**; a strip of fabric **308(a)**, **308(b)**; and an interlocking component **312(a)**, **312(b)**. The interlocking system **300** of FIG. 3 is a combination of two complementary interlocking devices. Although FIG. 3 depicts the sleeve **304(a)**, **304(b)** and the sleeve engagement piece **306(a)**, **306(b)** as having a hollow tube-shape with a triangular cross-section, it is contemplated that the sleeve **304(a)**, **304(b)** may have a cylinder shape or any hollow tube-shape. It is also contemplated that the sleeve engagement piece **306(a)**, **306(b)** may have a corresponding cross-sectional shape, where corresponding shape may mean either the same shape at the perimeter, or a different shape so long as it fulfills the purpose of engagement and securement of the material M within the sleeve **304(a)**, **304(b)**, respectively. Thus, the cross-sectional shape of the engagement piece **306(a)**, **306(b)** corresponds to the cross-section of the sleeve **304(a)**, **304(b)**, respectively, in the sense that the engagement piece **306(a)**, **306(b)** fits therein and maintains the material in place. For example, if the sleeve **304(a)**, **304(b)** has a hollow tube shape with a triangular cross-section, the sleeve engagement piece **306(a)**, **306(b)** may have a correspondingly sized triangular cross-section shape as well to be accommodated therein, as shown in FIG. 3.

[0038] In addition, the first sleeve engagement piece **306(a)** may be proportionally smaller than the first sleeve **304(a)** such that the first sleeve engagement piece **306(a)**, when inserted alone, loosely fits within the first sleeve **304(a)**. In this type of embodiment, a portion of material M is to be wrapped around the first sleeve engagement piece **306(a)** before the first sleeve engagement piece **306(a)** is inserted into the first sleeve **304(a)**. The first sleeve engagement piece **306(a)** may be proportionally sized in order to allow the first sleeve engagement piece **306(a)** to fit snugly in the first sleeve **304(a)** when the first sleeve engagement piece **306(a)** is first wrapped with a portion of material M.

[0039] It is contemplated that material M may vary in thickness, pliability, and density. Accordingly, it is contemplated that alternate sleeve engagement pieces may be necessary. FIG. 3 depicts sleeve engagement piece 306(c) as being a complementary shape to first sleeve 304(a) and as proportionally larger than first sleeve engagement device 304(a). In certain situations, it may be necessary for the first track 302(a) to utilize sleeve engagement piece 306(c) instead of first sleeve engagement piece 306(a). For example, in situations where material M is so thin that when wrapped around the first sleeve engagement piece 306(a), the wrapped sleeve engagement piece 306(a) does not snugly fit in sleeve 304(a). In such circumstances, it may be necessary for the thin material M to be wrapped around a slightly larger sleeve engagement piece.

[0040] Although FIG. 3 depicts one alternate sleeve engagement piece 306(c) that is larger than the first sleeve engagement piece 306(a), it is contemplated that alternate sleeve engagement pieces could have various shapes, sizes, and textures. For example, in circumstances where material M is delicate, slippery, or has some other unusual characteristic. In such circumstances an alternate sleeve engagement piece could have a smooth texture with rounded edges to avoid puncturing a delicate material M, or an alternative sleeve engagement piece could have a roughly textured surface to increase grip on a slippery material. Installation of the track 302(a) may be in accordance with the description of FIG. 7.

[0041] FIG. 4 illustrates a front perspective view of an embodiment of a universal interlocking system 400 (hereinafter “interlocking system 400”). The interlocking system 400 may include a track 402 that may be similar to the track 102(a) and/or track 102(b) from FIG. 1A and may include similar components. Accordingly, the track 402 may include a sleeve 404, a strip of fabric 406, an interlocking component 408, and the sleeve 404 may include a wall gap 410.

[0042] FIG. 4 depicts track 402 partially installed on a damaged track 412. Damaged track 412 may include a damaged strip of fabric 414 and a damaged interlocking component 416. FIG. 4 illustrates that the wall gap 410 is wide enough to allow the damaged strip of fabric 414 connected to the damaged interlocking component 416 to freely or snugly pass through the wall gap 410, which causes the damaged interlocking component 416 to reside inside the sleeve 404. Although the damaged interlocking component 416 is herein depicted as a replacement row of teeth, it is contemplated that replacement interlocking components (i.e., interlocking component 408, etc.) may replace a damaged row of teeth from a zipper system and replace the damaged teeth with a different interlocking component. For example, a damaged zipper track may be replaced with a track containing a Velcro component or other interlocking closure device/system type.

[0043] FIG. 5 illustrates a front perspective view of an embodiment of a universal interlocking system 500 (hereinafter “interlocking system 500”) depicted in this embodiment. The interlocking system 500 may include a track 502 that is configured to attach to any compatible surface. The track 502 includes a surface anchor component 504, a strip of fabric 506, and an interlocking component 508. It is contemplated that the surface anchor component 504 may be chosen according to the surface to which it is intended to be attached. For example, compatible anchor components (e.g., adhesive(s), magnets, elastic suction cups, Velcro, etc.) would be selected to attach the interlocking system 500 to the surface of a desired location (building wall, room wall, ceiling, surface adjacent an area desired for blocking off or enclosing) using at least one of: adhesion, magnetism, suction, etc. While FIG. 5 depicts the surface anchor component 504 as a generic two-dimensional, tubular shape, it is contemplated that the surface anchor component may be of whatever shape the chosen anchor component has (e.g., elongated planar shapes, such as a tape or ribbon, etc.) for the attachment method desired and/or that it is conformable to the shape of the desired location on which the surface anchor component is to be placed.

[0044] Notably, the embodiment of FIG. 5, and all embodiments, may incorporate one or more features from the other embodiments disclosed herein. For example, the track 502 may incorporate circuit elements like those discussed with respect to FIG. 3, since it is conceivable that one may want to highlight the location of the entrance or add a power or data line through the closure

system.

[0045] Additionally, it is contemplated that the interlocking component **508** may be similar to traditional interlocking zipper teeth, hook and loop structures, Ziploc®, or other interleaved binding elements/configurations.

[0046] FIG. **6** illustrates a top-front perspective view of an embodiment of a universal interlocking closure system **600** (hereinafter “interlocking system **600**”) and an end view of interlocking system **600**, each of which are similar to the embodiment shown in FIG. **1A**, except that system **600** may further include a wall gap **602** that includes one or more protrusions **604** that protrude intermittently from one or both faces of the wall gap **602** toward the opposite face of the wall gap **602**, along a length of the wall gap **602**. The protrusions **604** may facilitate a tighter hold upon the interlocking system being replaced upon installation by biting into the system being replaced. Although FIG. **6** depicts the one or more protrusions **604** as triangles, it is contemplated that the one or more protrusions may be any suitable shape.

[0047] FIG. **7** illustrates a method **700** of installing an interlocking closure device/system, like those described above. The order in which the operations or steps are described is not intended to be construed as a limitation, and any number of the described operations may be combined in any order and/or in parallel to implement method **700**.

[0048] At step **702**, the method **700** may include determining the desired configuration for the interlocking component to be used in replacement or addition. For example, if replacing a damaged zipper-style track with a functioning zipper-style track, a user may determine to install a replacement zipper track. In the event a user is replacing a damaged hook-and-loop-style track with a new hook-and-loop-style track, if one side is all loops and the other side is all hooks, the user may determine whether to install both sides of the system using two corresponding tracks for a completely new hook-and-loop-style system; or the user may determine to only replace one side that is damaged. In such a situation, a further determination is made to install the corresponding hook-side of a track or the loop-side of a track.

[0049] It is noted that other factors may be considered when determining which materials should be used for the track being installed. For example, if a new system is being installed in an environment (as compared to a wearable item or a consumer product such as a bag, tent, etc.) a track made of certain plastics may not be suitable for use in high-heat environments and/or certain metallic components may not be advantageous in environments saturated with combustible fumes.

[0050] The method **700** may further include a step of determining the desired location for the replacement track to be installed. For example, if replacing a zipper style track on a garment, the location for the replacement track is determined by the location of the damaged track. However, if the replacement track is being installed in a location that did not previously have a track installed, then determining the desired location may require multiple considerations including ergonomic considerations and spatial considerations.

[0051] At step **706**, the method **700** may further include determining the desired coupling method required to couple the sleeve of the new track to the installation item or location. For example, if the location for track installation is plastic sheeting to create a temporary passage into an environment that is best left enclosed when possible (e.g., a paint room, an area being debugged with fogging, etc.), an embodiment of a track with a sleeve that is coupled by applying heat to cause the sleeve to shrink may damage the plastic sheeting. Additionally, if an embodiment of a track utilizing an anchor component is to be installed, then the installation surface may be important. For example, if the surface is a steel beam with an uneven surface, then an anchor component configured with suction cups may not be effective, but an anchor component configured with one or more magnets may be effective. Moreover, if an embodiment of a track that required the use of a sleeve engagement piece is to be installed, then the characteristics of the type of material where the track is to be installed may be important. For example, if the coupling material is thin and delicate, then a larger sleeve engagement piece without sharp edges may be needed.

However, if the coupling material is thick and inherently slippery, then a smaller engagement piece with a textured surface or small protrusions extending along the surface may be necessary to get a grip on the coupling material.

[0052] At step **708**, method **700** may include determining if a track being installed is replacing a previously installed track. For example, if the new track is being installed on a jacket with a damaged zipper-style track, then the new track may be replacing an existing track. If the new track is being installed to replace a previously installed track, method **700** may continue to step **710**. If the new track is not being installed to replace a previously installed track, method **700** may continue to step **718**.

[0053] Notably, it is contemplated that a new track may be installed to replace existing tracks and/or interlocking systems that are not damaged. For example, the owner of a jacket having a fully functioning zipper-style system may desire a jacket with a hook-and-loop style system.

Additionally, or alternatively, the owner of a jacket having a fully functioning zipper may desire a jacket with a zipper-style interlocking system that has lighting components, portable device power source wires, or audio wires embedded into the interlocking system.

[0054] At step **710**, method **700** may include aligning an end (i.e., a top end or bottom end) of the sleeve of the track to be installed with an end of the interlocking mechanism that is being replaced and ensuring that the wall gap within the sleeve of the track to be installed is aligned with the strip of material coupled to the interlocking mechanism being replaced.

[0055] At step **712**, method **700** may include positioning the interlocking mechanism being replaced into the sleeve of the track that is being installed such that the entirety of the interlocking mechanism being replaced is fully within the sleeve of the track being installed (i.e., sliding the replacement track onto the track being replaced).

[0056] If the coupling method for the sleeve of the track being installed requires heat to be applied to the sleeve, then the method **700** may continue to step **714**. If the coupling method for the sleeve for the sleeve of the track being installed requires plastic deformation of the sleeve, then the method **700** may continue to step **716**.

[0057] At step **714**, the method **700** may include applying adequate heat to the sleeve of the replacement track such that the sleeve constricts around the interlocking component being replaced and the new track is adequately coupled.

[0058] At step **716**, the method **700** may include deforming the sleeve of the placement track such that the sleeve compresses around the interlocking component being replaced and the new track is adequately coupled.

[0059] At step **718**, the method **700** may include determining if a sleeve engagement piece is required to install the new track. For example, if an embodiment of a new track similar to the embodiment in FIG. 3 is being installed on plastic sheeting to create an opening that could be sealed and resealed with an interlocking closure system, installation of the embodiment would require the use of a sleeve engagement device. If a sleeve engagement piece is necessary, the method **700** may continue to step **720**. Additionally, if an embodiment of a new track being installed is similar to the embodiment in FIG. 5, the replacement track would not require the use of a sleeve engagement piece to be installed. If a sleeve engagement piece is not necessary, the method **700** may continue to step **724**.

[0060] At step **720**, the method **700** may include wrapping the material from the installation location around the sleeve engagement piece. For example, if the replacement track requiring a sleeve engagement piece is being installed on plastic sheeting, then the portion of the plastic sheeting where the track is to be installed may be wrapped around the sleeve engagement piece such that the sleeve engagement piece is completely or partially enveloped by the plastic sheeting.

[0061] At step **722**, the method **700** may include positioning the wrapped sleeve engagement piece into the corresponding sleeve. For example, the sleeve may slide onto the wrapped sleeve engagement piece. It is also contemplated that the wrapped sleeve engagement piece may be

inserted into the sleeve by forcing (e.g., pushing, snapping, etc.) the wrapped sleeve engagement piece through the gap wall within the sleeve.

[0062] In contrast, at step **714**, the method **700** may include preparing the surface of the installation location to couple with an anchor component. For example, if an embodiment of a track similar to the embodiment depicted in FIG. 5 has an anchor component configured to couple to a surface using an adhesive, the mounting surface may be cleaned and prepared for adhesion with the anchor component. Additionally, if an embodiment of a track similar to the embodiment depicted in FIG. 5 has an anchor component configured to couple to a surface using suction cups, the mounting surface may be prepared to provide a smooth surface for coupling with suction cups (i.e., sanded, polished, free of dust, etc.).

[0063] At step **726**, the method **700** may include coupling the anchor component to the surface of the location where the new track is to be installed. It is contemplated that coupling the anchor component to the surface may require the use of adhesion, magnetism, suction, tension, pressure, or force.

[0064] Additional miscellaneous steps that may be included with the method or considered as part of the system componentry (as consistent with the above) may include: connecting the circuit elements to one or more of a heating controller, a lighting controller, a battery, an audio output/input device (e.g., headphones, phone, speakers, digital audio player, etc.); controlling a lighting scheme of the enclosure system; cutting to size the tracks depending on the item/location of installation; controlling the heating of a separately connected item, such as a coat, gloves, hat, etc.; or a combination of the aforementioned items.

Conclusion

[0065] Although several embodiments have been described in language specific to structural features and/or methodological acts, it is to be understood that the claims are not necessarily limited to the specific features or acts described. Rather, the specific features and acts are disclosed as illustrative forms of implementing the claimed subject matter.

Claims

1. An interlocking device, comprising: a track including: a sleeve, a gap extending longitudinally along a first side of the sleeve, an interlocking component extending longitudinally along a second side of the sleeve, the second side being opposite the first side.
2. The interlocking device of claim 1, wherein: the track is a first track; the sleeve is a first sleeve; the gap is a first gap; the interlocking component is a first interlocking component; and further comprising: a second track that is engageable with the first track, the second track including: a second sleeve, a second gap extending longitudinally along a first side of the second sleeve, and a second interlocking component extending longitudinally along a second side of the second sleeve, the second side of the second sleeve being opposite the first side of the second sleeve, and the second interlocking component configured to engage with the first interlocking component.
3. The interlocking device of claim 2, further comprising: a sleeve engagement piece, the sleeve engagement piece configured to couple with the sleeve, and a third track, the third track comprising: a third interlocking component that is configured to engage with at least one of the first interlocking component or the second interlocking component; and a surface anchor component, wherein a first side of the surface anchor component is coupled to the third interlocking component and the second side of the surface anchor component is coupled to a surface, the second side of the surface anchor component being opposite the first side of the surface anchor component.
4. The interlocking device of claim 3, wherein the surface anchor component is coupled to the surface utilizing at least one of adhesion, magnetism, suction, tension, pressure, or force.
5. The interlocking device of claim 1, wherein the interlocking component and the second side of the sleeve are separated by a strip of material, the material extending an entire length of the

interlocking component and the entire length of the second side of the sleeve.

6. The interlocking device of claim 5 wherein the strip of material contains part of a circuit.

7. The interlocking device of claim 6 wherein the part of a circuit is one of one or more optical fibers or one or more electrical conductors.

8. The interlocking device of claim 1 wherein the interlocking component is one of a row of teeth, a continuous interlocking structure, a Velcro component, magnets, buttons, clasps, or other suitable fastener.

9. An interlocking system comprising: a track including: a sleeve, a gap extending longitudinally along a first side of the sleeve, a first edge of a strip of material adhered to a second side of the sleeve and extending longitudinally along the second side of the sleeve, the second side of the sleeve being opposite the first side of the sleeve, and an interlocking component adhered to a second edge of the strip of material and extending longitudinally along the second edge of the strip of material.

10. The interlocking system of claim 9, wherein: the track is a first track; the sleeve is a first sleeve; the gap is a first gap; the strip of material is a first strip of material; the interlocking component is a first interlocking component; and including: a second track that is engageable with the first track, the second track including: a second sleeve, a second gap extending longitudinally along a first side of the sleeve, a first edge of a second strip of material adhered to a second side of the second sleeve and extending longitudinally along the second side of the second sleeve, the second side of the second sleeve being opposite the first side of the second sleeve, and a second interlocking component adhered to a second edge of the second strip of material and extending longitudinally along the second edge of the second strip of material, the second interlocking component configured to engage with the first interlocking component.

11. The interlocking system of claim 9 wherein the strip of material contains part of a circuit.

12. The interlocking system of claim 11 wherein the part of a circuit is one of optical fibers or electrical conductors.

13. The interlocking system of claim 12 wherein the part of a circuit is configured to couple with one or more circuit interaction components.

14. The interlocking system of claim 9 wherein the interlocking component is one of a row of teeth, a continuous interlocking structure, a Velcro component, magnets, buttons, clasps, or other suitable fastener.

15. The interlocking system of claim 14 wherein the interlocking component is configured to engage with a second interlocking component upon longitudinal manipulation of a sliding joiner piece.

16. A method for installing an interlocking system comprising: providing a track, the track including: a sleeve, a gap extending longitudinally along a first side of the sleeve, a first edge of a strip of material adhered to a second side of the sleeve and extending longitudinally along the second side of the sleeve, the second side of the sleeve being opposite the first side of the sleeve, an interlocking component adhered to a second edge of the strip of material and extending longitudinally along the second edge of the strip of material; positioning the track to a desired location, wherein the desired location is where an interlocking system is to be installed; and coupling the track to the desired location.

17. The method of claim 16, wherein: applying heat above a threshold causes the sleeve to constrict; and constricting the sleeve couples the track to the desired location.

18. The method of claim 16, wherein plastically deforming the sleeve couples the sleeve to the desired location.

19. The method of claim 16, wherein the track further includes: a sleeve engagement piece, the sleeve engagement piece configured to couple with the sleeve; and wherein the method further comprises: enveloping the sleeve engagement piece with material at the desired location; and positioning the sleeve around the enveloped sleeve engagement piece.

20. The method of claim 16, wherein: the track is a first track; the interlocking component is a first interlocking component; and further comprising: a sleeve engagement piece, the sleeve engagement piece configured to couple with the sleeve, and a second track, the second track comprising: a second interlocking component that is configured to engage with the first interlocking component; and a surface anchor component, wherein a first side of the surface anchor component is coupled to the second interlocking component and a second side of the surface anchor component is coupled to a surface, the second side of the surface anchor component being opposite the first side of the surface anchor component.
